Doncaster Property Model: how it works and what it's for

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Summary

The Doncaster Property Model estimates what any home in Doncaster would sell for today, says how sure it is, and turns that into investment numbers. It is built only from open public data, and it runs in an interactive dashboard in the browser.

On 5,995 sales from January 2025 to August 2026 that it never saw, its median error is **8.1%**, and 59% of estimates fall within 10% of the sale price. The method behind online "worth now" calculators (last sale price times the house price index) scores 13.0% on the same homes.

Around the valuation sit four further tools:

- a rent estimate and a full buy-to-let deal analyser;
- a buy, refurbish and sell calculator;
- returns, growth and risk for Doncaster's 39 neighbourhoods;
- a market view with price indices and tenant-risk measures.

It is meant for anyone checking an asking price, screening a buy-to-let or choosing where to buy in Doncaster.

Aims

The model is built to answer seven practical questions about a Doncaster home or area:

Question	Where in the dashboard	What it gives
What is this home worth today?	Value a property	Estimate, 80% range, a confidence level, comparable sales and what drove the figure
Is the asking price fair?	Value a property, Investing	Asking price against the estimate and range, and against recent sales in its neighbourhood

Question	Where in the dashboard	What it gives
What would it rent for?	Investing	A rent estimate with its likely range, the bedroom average and the housing benefit cap
Does it work as a buy-to-let?	Investing	Stamp duty, mortgage, tax as an individual or a company, a lender stress test, 10-year projection and IRR
Would works pay for themselves?	Value a property, Investing	Value added by an extension, loft, energy upgrade or refurbishment, against sourced 2026 costs
Which neighbourhoods suit buy-to-let?	Areas and rents	Yield, growth and total return against what the same cash earns elsewhere, after a risk allowance
How have prices moved, and what are the risks?	Areas and rents, Market	Price indices since 1995, the 2008 fall, landlord court claims, anti-social behaviour, flood and energy rules

Three design aims run through all of it:

- **Judged the way it is used.** The model is tested on later sales than it learned from, and it never uses information that would not have been known at the time.
- **Honest about uncertainty.** Every estimate carries a range that has been checked, and a confidence level based on how well similar homes were valued.
- **Assumptions in the open.** Every cost and rate has a source and can be changed on the page.

Data

Everything comes from free public sources, joined at the level of the individual home where possible. The valuation learns from 50,042 standard Doncaster sales since 2015.

Source	What it gives the model	Coverage
HM Land Registry Price Paid Data	Sale prices, dates, type, tenure, new builds, each home's earlier sales	Doncaster, 1995 to August 2026

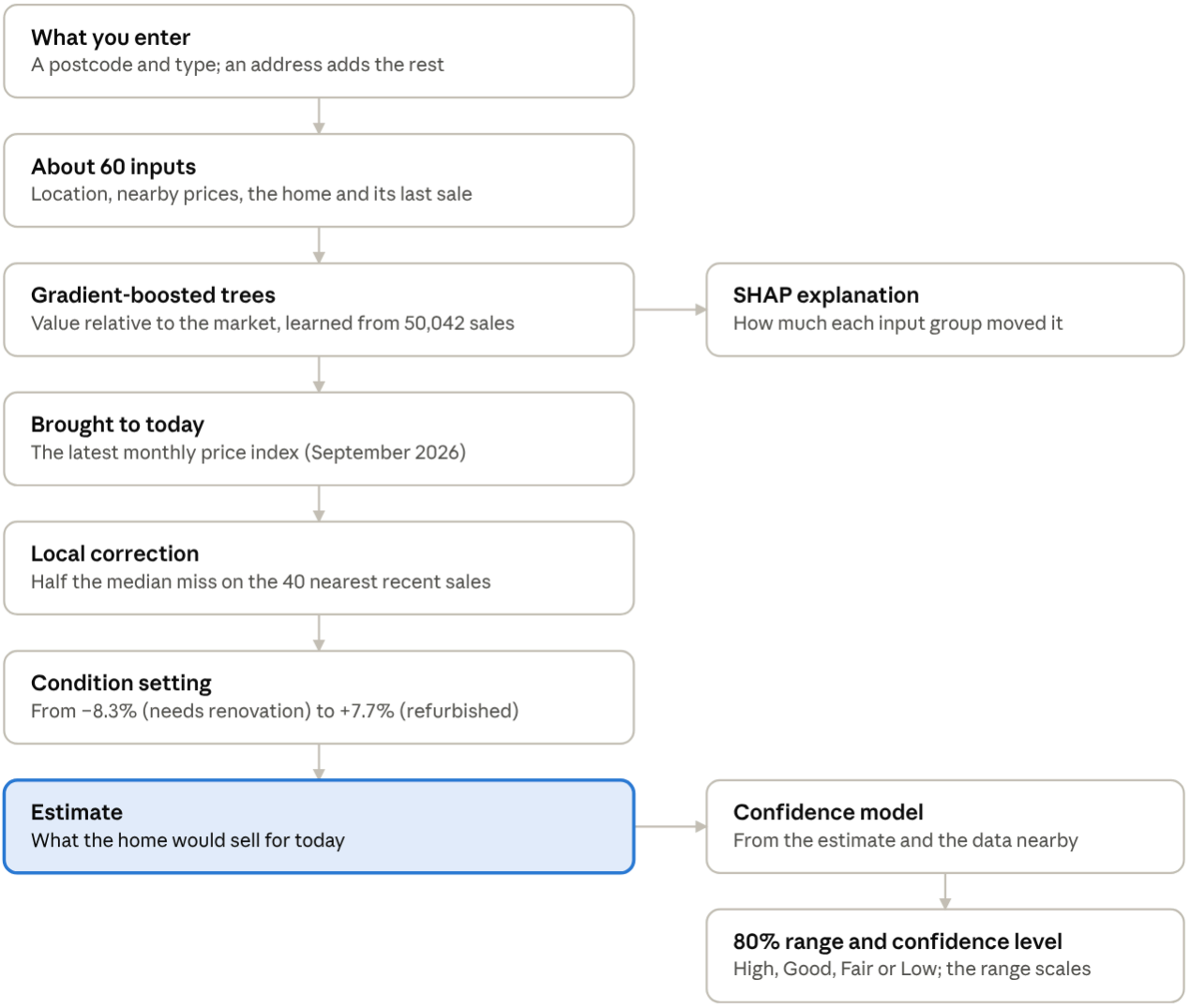
Source	What it gives the model	Coverage
Energy Performance Certificates (MHCLG)	Floor area, rooms, build age, energy score, walls, loft, glazing, heating, why and when each certificate was done	130,780 Doncaster certificates, 2012 to September 2026
HM Land Registry title plans with OS Open UPRN	Plot size for each home	Doncaster
postcodes.io (ONS Postcode Directory)	Coordinates and 2021 neighbourhood codes	All Doncaster postcodes
English Indices of Deprivation 2025	Income, employment, education, health, crime, housing and environment scores	199 small areas (LSOAs)
ONS rents (Price Index of Private Rents; private rental market statistics)	Doncaster rents by type and bedrooms; the spread of rents	August 2026; quartiles from October 2022 to September 2023
UK House Price Index	Prices by type for the rent model; a check on the model's own index	Council areas in England and Wales
DWP Local Housing Allowance	Housing benefit caps	2026-27
Census 2021	Tenure (owned, social and private rented) and residents	Doncaster small areas
Ministry of Justice possession statistics	Landlord court claims and bailiff evictions	Quarterly to April to June 2026
police.uk	Anti-social behaviour reports, placed on the map	6,317 reports, September 2025 to August 2026
Environment Agency, NaPTAN, DfE and Ofsted, OpenStreetMap, OS building outlines, VOA council tax	Flood risk, transport, schools, local amenities, building size, council tax bands	Shown for each home; tested in the model
Cost guides and official guidance	Building and refurbishment costs; stamp duty, tax, energy and rental rules	2026, each with a source in the dashboard

Raw data is downloaded by a script rather than stored with the code. The energy certificate file needs a free GOV.UK One Login.

The valuation model

An estimate comes from gradient-boosted decision trees (LightGBM) trained on Doncaster's own sales, then corrected for local trends and adjusted for condition. Only a postcode and a property type are required; every other input improves the estimate when it is known.

Six steps lead to each estimate, which is then explained and given a range



How an estimate is made · 6 steps and 2 side models

The main column runs top to bottom; the explanation reads the trees, and the confidence model reads the finished estimate.

Prices on a common footing. A monthly quality-adjusted price index (a hedonic time-dummy model, January 2015 = 1) takes the market's ups and downs out of every sale. The trees learn what a home is worth relative to the market, and the latest index brings that to today's prices (September 2026).

What the trees see. About 60 inputs, in groups:

- **Location and deprivation:** coordinates, distance to the centre, and the small area's overall deprivation score with its seven domains.
- **Prices nearby:** typical levels for the postcode sector, small area, street and postcode, and the 10 nearest earlier sales.
- **The home itself:** floor area, rooms, plot size, type, built form, age, tenure, new build, and energy score and potential.
- **Construction detail:** walls, insulation, loft, glazing, heating, fireplaces and ceiling height.
- **Its certificate history:** why the latest certificate was done, whether it was let, and how its energy score changed.
- **Its last sale:** the price, brought to today with the index, and how long ago it was. This is the strongest single input: estimates are 7.6% out with it and 9.4% without.

A local correction. One Doncaster-wide index under-values areas rising faster than the rest. Each estimate is shifted by half of the model's median miss on the 40 nearest sales of the last 12 months, which cut the spread of area-level bias from 2.3% to 1.5%.

Condition. No open dataset records condition, so it is a setting, calibrated on 1,007 homes resold within 3 to 18 months, which are mostly bought, refurbished and resold. At purchase they sold 7.6% below a model without last-sale information, and at resale 7.7% above it. Homes resold after 5 years or more sold about 1.4% above it at both ends, for reasons that aren't a quick refurbishment, so that is taken off: −8.9% for "needs full renovation" and +6.2% for "recently refurbished". Each home's sales stay in one test fold. "Dated" (−4.5%) and "good" (+3.1%) sit halfway. When the estimate already knows a last sale, the model has partly priced condition in, so the setting is smaller: −4.7% for "needs full renovation", and +0.6% for a resale within 18 months of that sale.

A range and a confidence level. Each estimate has an 80% range, set from how far past estimates missed for homes in the same price band, with or without a known last sale. A second, small model predicts how far off this particular estimate is likely to be, from things like how many sales are nearby and how old the certificate is. That gives a confidence level (High, Good, Fair or Low), and the range widens or narrows with it.

An explanation. A SHAP breakdown, computed in the browser, shows how much each group of inputs moved the value up or down from a typical Doncaster sale. It matches LightGBM's own figures to 9 decimal places.

What was tested and left out. Flood risk, transport, schools, amenities, urban or rural class, building footprints, exact address positions and the council tax mix were each tested over three random seeds. None beat seed-to-seed noise, so they are shown for each home but do not move the estimate.

How accurate it is

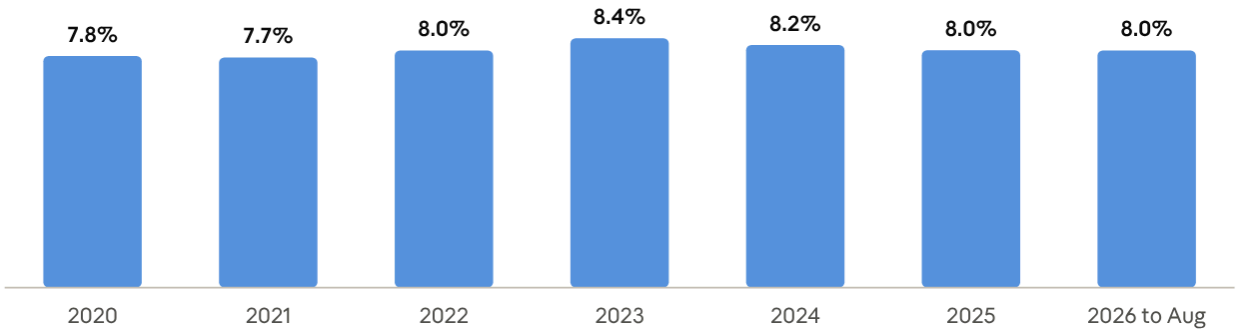
The model was tested the way it would be used: trained on 2015 to 2023 sales, tuned on 2024, then scored on the 5,995 sales from January 2025 to August 2026. Local price inputs and the index use only sales published at least two months before, and energy ratings only certificates lodged before the sale. Not everything is strictly dated. Size and build details can come from a certificate lodged after the sale when there is no earlier one, which affects 35 test sales (0.6%). A strict re-run allowing only certificates lodged by the sale date, over three random seeds, changes the overall median error by −0.03 points: little aggregate change, though those 35 homes are valued less well (9.5% to 15.5%). Deprivation scores and plot sizes apply to every sale whatever its date, and neighbour price per m² uses each neighbour's best-known floor area; dropping the first two is only a sensitivity test (+0.24 points).

Land Registry registers some sales months late, new builds most of all: they are 4.3% of the 2025 test sales and none from February 2026. New builds are the easiest homes to value (5.3% median error), and January to August 2026 scores 8.1%, much the same as those months of 2025, so the lag doesn't flatter the result. It will be scored again once the register catches up.

Result on the test sales	Value
Median error	8.1%
Within 10% of the sale price	59%
Within 20% of the sale price	84%
Mean absolute error	£22,813
Sale prices inside the 80% range	80.5%

Median error stayed between 7.7% and 8.4% in every year since 2020

Each year valued by a model trained only on sales up to two years before, with the price index as known then



Year-by-year backtest, 2020 to August 2026 · the model's own test results

The result holds from year to year: the worst year, 2023, was 8.4%. Each data layer was kept only if it beat seed-to-seed noise. Location and deprivation alone gave a 13.2% median error, energy certificate details took it to 9.7%, and the last sale to 8.7%. Plot size, street prices, tuning, the local correction and certificate history took it to 8.0% with local prices from every earlier sale, or 8.1% under the two-month rule.

Against the quick methods, scored on the same test sales, with each method seeing only sales published at least two months earlier:

Method	Share of homes it can value	Median error	Within 10%	This model on the same homes
Last sale × UK House Price Index for Doncaster (online "worth now" calculators)	72%	13.0%	41%	7.6%
Last sale × UK House Price Index for the property type	72%	12.7%	42%	7.6%
Median £/m² of the 10 nearest same-type sales × floor area	98%	13.6%	39%	8.0%
Median £/m² of same-type sales in the postcode sector × floor area	96%	14.9%	36%	7.9%
Median price of same-type sales in the postcode sector	99%	17.5%	31%	8.1%

Method	Share of homes it can value	Median error	Within 10%	This model on the same homes
Hedonic regression on the model's own features, with a fixed effect per postcode sector	100%	9.8%	51%	8.1%

The last-sale method gets worse as the last sale ages: 8.6% when it was under 5 years ago and 18.1% at 10 years or more, against the model's 6.2% and 8.6% on the same homes.

Confidence levels work. Homes rated High confidence (7% of homes) had a 4.9% median error, Good (29%) 6.6%, Fair (33%) 8.3% and Low (31%) 11.3%. Each level's 80% range held the price for 76% to 81% of its sales.

Fairness checks. The assessors' ratio study (IAAO) gives a uniformity score (COD) of 12.7, inside the standard of under 15. The price-related differential is 1.031, just over the 0.98 to 1.03 band, so dearer homes are valued slightly low relative to cheaper ones. Removing that bias cost 0.7 points of median error, so the estimates are left as they are.

Investing

The Investing tab takes the home's estimate and asks whether it works as an investment. It starts with the rent, then runs the full deal, a refurbishment and a comparison with the neighbourhood.

Rent estimate

No open data gives the rent of a single home, so rent is linked to value:

> **Rent** = Doncaster's average rent for the type × (the home's value ÷ Doncaster's average price for the type) ^ β

- **β = 0.596**, fitted across 316 council areas in England and Wales: ONS rents (August 2026) against UK House Price Index prices, with region and type effects.
- **Tested by leaving each area out:** its rents were predicted to within 5.9% of the actual rent (median), against 12.1% from the regional average alone. Inside Doncaster, rents follow prices across the four types with a slope of 0.48.
- **Likely range:** the middle half of Doncaster rents for the bedroom count (ONS, October 2022 to September 2023, the last edition with quartiles).
- **Not yet tested on individual lets**, because none are published, so the page asks for a check against listings.

Against the neighbourhood

The home is placed among the last three years of sales in its neighbourhood on three measures: price against estimated value, price per m², and gross yield at the modelled rent. Each is reported as the share of recent sales it beats, plus a combined score. Anti-social behaviour in the area and Doncaster's landlord court claims sit beside them.

Deal analyser

Part	What it does	Default
Purchase	Stamp duty with the additional-property surcharge, legal and survey costs	5% surcharge, £1,500
Mortgage	Deposit, rate, term and arrangement fee	25%, 5.66% (Moneyfacts, August 2026), 25 years, £1,999
Lender stress test	The rent a lender needs at its stress rate	5.5%; cover of 125% (basic rate or company) or 145% (higher rate)
Running costs	Letting and management, maintenance, voids, insurance, safety checks	12% and 8% of rent, 2 weeks, £300, £250 a year
Tax	As an individual (Section 24: a basic-rate credit on the interest or the property profit, whichever is lower, with the rest and any loss carried forward) or a limited company (corporation tax from 19% to 25%, with marginal relief)	2026-27 rates, and the higher property income rates from April 2027
Energy rules	The cost of reaching EPC C, capped as the rules do	£5,400 average, £10,000 cap
Projection	Equity, cumulative cash and an IRR over 10 years after tax on the gain when sold (capital gains tax above £3,000: 18% on the part within the unused basic-rate band, 24% above it; or corporation tax. Only improvements come off the gain), plus stress scenarios	Past Doncaster rent growth; the neighbourhood price growth the scorecard uses

Part	What it does	Default
Mortgage rate	The first year's monthly cash flow before and after tax at every rate, with the break-even rate and where the stress test fails; a slider sets the deal's rate	0% to 12%, wider for a higher rate
What matters most	How far the 10-year IRR, or the first year's cash flow, moves when each input is lowered and raised, one at a time	Price paid and rent ±10%; rate and growth rates ±1 point; voids ±2 weeks; maintenance ±50%; deposit ±10 points, not below 25%
Past decades	The deal's equity and after-tax IRR if prices had followed the neighbourhood's index over the 10 years after each purchase year; overlapping stretches, so it shows how much timing has mattered, not a forecast	22 stretches, 1995 to 2026, against the 10-year gilt
Comparison	Total return against what the same cash earns elsewhere, after a risk allowance	10-year gilt (5.48%); Bank Rate (3.75%) or your own rate

Works and refurbishment

The What-if card re-runs the model with an extension, loft conversion, better energy rating or refurbishment, and sets the value added against a 2026 cost. Works lift a home's condition only if it starts no worse than the condition they put right, so decorating a home that needs a full renovation adds nothing for condition. The buy, refurbish and sell calculator buys at your price (or, without one, the model's value in the condition the works put right) and sells at the value after them. It finances 70% of the price as a bridging loan at 12% a year with a 2% fee, and gives the highest price that breaks even and the highest for a 15% margin on the sale price.

Refurbishment costs are Yorkshire guide prices (less.co.uk, April 2026), item by item, with a 10% contingency (Federation of Master Builders):

- **Fixed amounts:** a bathroom (£2,300 to £5,800 each) and a boiler (£1,950).
- **Grow with the square root of floor area:** a kitchen (£4,550 to £10,000 for a typical 3-bed semi) and a rewire (£5,940).
- **In proportion to floor area:** decorating (£37/m²), flooring (£32/m²), radiators (£26/m²) and re-skimming (£30/m²).

- **Finish follows value:** kitchens and bathrooms run from the bottom of the budget range for cheap homes to mid-range for the dearest, unless a finish is chosen.

For a £228,000 semi of 85 m², a new kitchen, bathroom, decoration and flooring comes to about £19,300. The model expects them to add about £17,600 of value, so the works roughly pay for themselves before buying and selling costs. A refurbishment only makes money when the home is bought below its "dated" value.

Areas

The Areas tab compares Doncaster's 39 neighbourhoods on price growth, rental return and risk, so you can shortlist where to buy before looking at any one home. The neighbourhoods are the 2021 MSOAs, named by the House of Commons Library, and each holds about five of the map's 199 small areas.

Price growth since 1995

- **Method:** a repeat-sales index (Bailey-Muth-Nourse with Case-Shiller weights) on 50,488 pairs of sales of the same home since 1995. Because it compares each home with itself, a change in the mix of homes sold doesn't move it.
- **Neighbourhoods:** each index is Doncaster's plus a smooth local deviation. The amount of smoothing was chosen by cross-validation on held-out pairs, and it beat Doncaster's index alone. 80% ranges come from 200 bootstrap resamples.
- **Checked against the official index:** Doncaster's annual changes correlate 0.986 with the UK House Price Index, and its 10-year growth to 2025 is 4.49% a year against the official 4.50%.
- **Time-lapse:** pick any period with a slider or quick buttons. The map colours each neighbourhood by its growth against Doncaster's, and a bar race ranks them. From 2007 to 2013, Doncaster fell 16% and all 39 neighbourhoods fell, from −5% in Tickhill & Wadworth to −24% in Central Doncaster & Hyde Park.

Return on a typical home

Each neighbourhood's return is worked out for a cash purchase at its median price, let at Doncaster's ONS rent for the type:

> **Return above the alternative** = net yield + price growth a year − risk allowance − opportunity cost

Net yield takes off the deal analyser's running costs and spreads stamp duty, legal and selling costs over a 10-year hold. Price growth is the average of the past 10 years (or 5), not a forecast. The opportunity cost is the 10-year gilt by default (5.48% at the close on 8 October 2026), with Bank Rate or your own rate as alternatives.

Risk allowance

The allowance is on from the start and takes 0.5 to 1.9 points a year off each neighbourhood's return. A switch turns it off and every setting can be changed.

Allowance	How it is worked out	Default
Tenants	Rent × the cost of rent guarantee insurance × Doncaster's landlord court claims against England's × the area's risk index (its anti-social behaviour and deprivation against Doncaster's)	3.75% of rent, × 0.79, index kept between 0.4 and 3
Flood	Share of homes at flood risk × extra insurance	£445 a year, a scenario to check with a quote: a typical let falls outside Flood Re, and before Flood Re high-risk homes with no claims were quoted 148% above the market (Flood Re, £524 against £211 in 2016)
Energy rules	Share of homes below EPC C × the cost to reach C, spread over the hold	£5,400, capped as the rules do
Downturn	The area's own 2008 fall, once in every 30 years	One downturn in 31 years of data

Best buy-to-let neighbourhoods

A ranked list, an income-against-growth chart and a sortable table cover all 39, with minimum and maximum price filters. After the risk allowance, the top five are Mexborough West, Moorends, Edlington, Conisbrough South and Central Doncaster & Hyde Park. The map colours any measure red for worse and blue for better, with grey near the Doncaster average.

Market and tenant risk

The Market tab tracks Doncaster as a whole: prices up 61% from 2015 to 2026 on the repeat-sales index, with the cheaper areas rising fastest.

- **Doncaster house price index:** quality-adjusted each month, so the same mix of types and neighbourhoods is compared over time.
- **Prices by area tier:** postcode sectors split into thirds by their 2015-2019 prices. The cheapest third has risen fastest since 2015.
- **Median price by type and sales per month:** from Land Registry; the latest month is still filling in.

Tenancies ending in court. Ministry of Justice landlord possession claims, per 1,000 privately rented homes (Census 2021), over the latest four quarters:

Measure, year to June 2026	Doncaster	Yorkshire and The Humber	England
Standard landlord claims (include rent arrears)	4.6 (119 claims)	3.3	5.8

Doncaster's rate is higher than 53% of 309 English council areas. It also had 3.4 accelerated claims and 1.9 bailiff evictions per 1,000 rented homes. Accelerated claims were the paper route for no-fault Section 21 notices, which ended on 1 May 2026. These figures exist only for the whole borough.

Anti-social behaviour. Every police.uk report in Doncaster from September 2025 to August 2026 (6,317) is placed in its small area by its map point and divided by residents. Doncaster's rate is 20.5 reports a year per 1,000 residents. The rates are mapped by neighbourhood and feed the tenant part of the risk allowance.

Around each home. The Value tab also shows flood risk, the nearest station, schools with their Ofsted grades, and the nearest supermarket, GP, park, motorway junction and main road.

Uses in practice

Five common jobs, and how the dashboard handles each:

1. **Check an asking price.** Paste a Rightmove or Zoopla listing, or pick the address so its energy certificate and last sale fill in. Compare the asking price with the estimate, its 80% range and the confidence level. An asking price above the top of a Good or High range is a reason to negotiate, and the comparable sales show what nearby homes went for.
2. **Screen a buy-to-let.** On the Investing tab, check the rent estimate against live listings, then run the deal with your own mortgage and tax position. Look at whether the lender stress test passes, whether the 10-year IRR beats the gilt, and how the home ranks against recent sales in its neighbourhood.

3. **Shortlist neighbourhoods.** On the Areas tab, set a price band, leave the risk allowance on, and compare the ranked list. Use the time-lapse to check how each area behaved over a period that matters to you, such as the 2007 to 2013 fall.
4. **Price a refurbishment or a flip.** Set the condition and the works, then compare the value added with the itemised cost. Put in a real quote when you have one. If the numbers only work at the guide's lowest finish, the margin is thin.
5. **Value a home in code.** `src/predict.py` runs the same model from the command line, with only a postcode and type required, and can print the SHAP breakdown.

In every case the estimate is a starting point for a decision, to be checked against a viewing, a survey and live listings.

Limits

The model is a screening tool, not a survey, a mortgage valuation or financial advice. The main places to be careful:

- **Condition is a guess you supply.** No open data records it, so a home's state comes from the setting, calibrated on quick resales. A home needing structural work can be worth far less than "needs full renovation" suggests.
- **Some estimates miss by a lot.** About 40% of estimates miss by more than 10%, and the Low confidence group has an 11.3% median error. The range and confidence level say how much weight to put on a figure.
- **Homes without a certificate are harder.** About 6% of sales have no matched energy certificate, so they are valued without a floor area and get a wider range.
- **Rents are borough-wide.** Area returns use Doncaster's average rent for each type, so cheaper neighbourhoods look better than they may be. The rent model has not been tested on individual lets.
- **The past isn't a forecast.** Growth figures are averages of past years, and the next 10 years may look nothing like the last.
- **Allowances, not measurements.** The risk allowance and the refurbishment finish that follows value are labelled assumptions. Guide costs don't say whether VAT is included.
- **Data lag.** Land Registry publishes sales about two months after they complete, so the latest months are thin. Court claims exist only for the whole borough, and anti-social behaviour depends on what gets reported.
- **Doncaster only.** The model is trained on Doncaster sales and shouldn't be used elsewhere.

How it's built and kept current

The model is a Python pipeline whose output is three things: a dashboard, a single-file public copy and a command-line tool.

- **Pipeline:** Python, with the joins in DuckDB SQL and the model in LightGBM. One entry point, pipeline.py, runs the scripts in order: download, build the dataset, match certificates, build the price index and features, train and test, then export. Each change was tested over three random seeds and kept only if it beat the noise between them. After testing, the model on the page is refitted on 46,566 kept sales from January 2015 to August 2025. Unit, browser, train-and-serve parity, privacy and clean-build tests ship with the code. Every assumption sits in one file with its source, and the public copy rebuilds from the zip byte for byte. A decision log records 138 choices and why they were made.
- **Dashboard:** eight tabs: Value a property, Investing, Areas and rents, Market, Model accuracy, Method, Saved and Changelog. The header shows the version number (21 on 10 October 2026), and the Changelog lists what changed in each version. The full model runs in the browser, with its 2,043 trees stored compactly in 6.9 MB, and agrees with the Python model to within 0.00000004 on the log scale. It has light and dark modes, and nothing animates when the reader's system asks for reduced motion.
- **Public copy:** one self-contained HTML file (about 11 MB) that runs in any browser with no server and can be hosted free on GitHub Pages. It leaves out house-level address data: the address lookup, sales history and saved list are hidden, and comparable sales show the street only. The shared code leaves out the address lookup too, since the EPC licence keeps certificate addresses under Royal Mail terms.
- **Command line:** `src/predict.py` values one home with the saved model, checks its inputs as the dashboard does, and can explain the estimate; a parity test checks it builds the same features as training.

Keeping it current. Land Registry publishes monthly: the August 2026 sales arrived on 29 September, and the next release is due around Oct 28, 2026. Each refresh downloads the new data, rebuilds the features, refits the model on the latest sales and re-exports the dashboard. ONS rents and police.uk reports also update monthly, and court claims quarterly. The Bank Rate alternative is next reviewed on Nov 5, 2026.